

FEASIBILITY REPORT



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INTRODUCTION

Conceptual study

- It represents the transformation of a project idea into a broad investment proposition, by using comparative methods of scope definition and cost estimating techniques to identify a potential investment opportunity. Capital and operating costs are usually approximate ratio estimates using historical data. It is intended primarily to highlight the principal investment aspects of a possible mining proposition.

Pre-feasibility study

- It is an intermediate-level exercise, normally not suitable for an investment decision. It has the objectives of determining whether the project concept justifies a detailed analysis by a feasibility study, and whether any aspects of the project are critical to its viability and necessitate in-depth investigation through functional or support studies. This study should be viewed as an intermediate stage between a relatively inexpensive conceptual study and a relatively expensive feasibility study.

Feasibility study

- This provides a definitive technical, environmental and commercial base for an investment decision. It uses iterative processes to optimize all critical elements of the project – **production capacity, technology, investment and production costs, sales revenues, and return on investment**. Normally it defines the scope of the work unequivocally, and serves as a base-line document for advancement of the project through subsequent phases.

THE ESSENTIAL FUNCTIONS OF THE FEASIBILITY REPORT ARE

- To provide a comprehensive framework of established and detailed exploits concerning the mineral project.
- To present an appropriate scheme of exploitation with designs and equipment lists taken to a degree of detail sufficient for accurate prediction of costs and results.
- To indicate to the project's owners and other interested parties the likely profitability of investment in the project it equipped and operated as the report specifies.
- To provide the information, in a form intelligible to the owner and suitable for presentation, to prospective partners or to sources of finance.

A FEASIBILITY REPORT BROADLY CONTAINS THE FOLLOWING INFORMATION

General:

- Topography, climate, population, access, services.
- Suitable sites for plant, dumps, townships, etc.

Geological:

- Structure, mineralization and genesis.
- Shape, size, type and extent of ore bodies, and extent of leached or oxidized ore.
- Assay data, assay compositing (ore zone and bench) data and statistical analysis.
- Rock properties, strength and stability.
- Mineral inventory (geological reserves) and mineable reserves, grade-tonnage curves and spatial distribution of grades.
- Associated waste rock volume.
- Ground water data.

Mining:

- Ultimate pit configuration, intermediate pit layouts and plans.
- Waste rock dilution and ore losses.
- Pre-production and development requirements.
- Ore production and waste stripping schedule.
- Waste disposal and dump formation schedule.

Metallurgy:

- Testing data of samples from drill cores and sludge samples.
- Type and stages of the extraction process.
- Results of small-scale pilot plant testing of bulk samples.
- Specification of degree of processing, and nature and quality of products.
- Effects of ore type and/or head grade variations upon the recovery and product quality.
- Treatment concept (in considerable detail), with flow-sheet and calculation of quantities flowing.
- Specification of recovery and product grade.
- General siting and layout of plant.

Ancillary services and requirements:

- Access, transport, power, water, fuel and communications.
- Workshops, offices, sundry buildings, equipment, etc.
- Labour structure and strength.
- Housing and transport of employees.
- Other social obligations.

Capital cost estimates

- Develop the mine and plant concepts and make all necessary drawings.
- Calculate or estimate the equipment list and all important quantities (of excavation, concrete, building area and volume, pipework, etc.).
- Determine a provisional construction schedule.
- Obtain quotes of the direct cost of items of machinery, establish the costs of materials and services, and of labor and installation.
- Determine the various and very substantial indirect costs, which include freight and taxes on equipment (may be included in direct), contractors' camps and overheads plus equipment rental, labor punitive and fringe costs, the owner's field office, supervision and travel, purchasing and design costs, licenses, fees, customs duties and sales taxes.
- Warehouse inventories.
- A contingency allowance for unforeseen adverse happenings and for unestimated small requirements that may arise.
 - - Operating capital sufficient to pay for running the mine until the first revenue is received.
 - - Financing costs and, if applicable, preproduction interest on borrowed money.

Operating cost estimates

- Define the labor strength, basic pay rates, fringe costs.
- Establish the quantities of important measurable supplies to be consumed - power, explosives, fuel, grinding steel, reagents, etc. - and their unit costs.
- Determine the hourly operating and maintenance costs for mobile equipment plus fair performance factors.
- Estimate the fixed administration costs and other overheads plus the irrecoverable elements of townsite and social costs.

Marketing

- Product specifications, transport, marketing regulations or restrictions.
- Market analysis and forecast of future prices.
- Likely purchasers.
- Costs for freight, further treatment and sales.
- Draft sales terms, preferably with a letter of intent.
- Merits of direct purchase as against toll treatment.
- Contract duration, provisions for amendment or cost escalation.
- Requirements for sampling, assaying and umpiring.

Rights, ownership and legal matters

- Mineral rights and tenure.
- Mining rights (if separated from mineral rights).
- Rents and royalties.
- Property acquisitions or securement by option or otherwise.
- Surface rights to land, water, rights-of-way, etc.
- Licenses and permits for construction as well as operation.
- Employment laws for local and expatriate employees separately if applicable.
- Agreements between partners in the enterprise.
- Legal features of tax, currency exchange and financial matters.
- Company incorporation.

Financial and tax matters

- Suggested organization of the enterprise, as corporation, joint-venture or partnership.
- Financing and obligations, particularly relating to interest and repayment on debt.
- Foreign exchange and reconversion rights, if applicable.
- Study of tax authorities and regimes, whether single or multiple.
- Depreciation allowances and tax rates.
- Tax concessions and the negotiating procedure for them.
- Appropriation and division of distributable profits

Revenue and profit analysis

- The mine and mill production schedules and the year-by-year output of products.
- Net revenue at the mine (at various product prices if desired) after deduction of transport, treatment and other realization charges.
- Calculation of annual costs from the production schedules and from unit operating costs derived previously.
- Calculation of complete cash flow schedules with depreciation, taxes, etc. for some appropriate number of years - individually for at least 10 years and grouped thereafter.
- Presentation of totals and summaries of results.
- Derived figures (rate of return, payback, profit split, etc.) as specified by owner or client.
- - Assessment of sensitivity to price changes and generally to variation in important input elements.

Environmental effects:

- Environment base line data.
- Environmental impact assessment.
- Environmental management planning.
- Environmental disaster management planning.
- Reclamation / mine closure planning.



THANKS